

CO4-AT2-Q2:

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Course Name: DAA for Complexity Analysis

Smart Waste Recycling Optimization (Sorting + Classification Algorithms)

Problem Understanding

A recycling facility processes different types of waste materials. Automated sorting systems are required to classify and organize waste efficiently for recycling.

Algorithms Used

Sorting Algorithm:

- Bubble Sort

Classification Concept:

- Material identification based on categories

Solution Design

The system:

- Collects waste material data.
- Classifies materials into groups.
- Sorts materials for processing.
- Improves recycling workflow.

Constraints

- Different material types
- Large volume of waste
- Processing speed requirements
- Accurate classification

Scalability

For large recycling plants:

- Machine learning classification can be added.
- Sensors and cameras can identify materials.
- Parallel sorting systems improve speed.

Performance Metrics

- Sorting accuracy
- Processing time
- Energy consumption
- Recycling efficiency

Result Analysis

Sorting and classification algorithms reduce manual effort and improve waste processing speed. Automated systems increase recycling efficiency and support environmental sustainability.

Conclusion

Algorithm-based waste management systems improve material recovery, reduce processing time, and help create efficient and sustainable recycling operations.